



SHOPEZ PLATFORM SUITE

PROJECT DESIGN PHASE

Solution Architecture Document

Data Models, API Endpoints, Middlewares & Tier Architecture Diagrams

PROJECT NAME:	ShopEZ Premium E-Commerce Platform
REPOSITORY NAME:	VIP-C2-ShopEZ-E-commerce-Application
PREPARED BY:	P Pranay
DATE:	June 2026
VERSION / STATUS:	v1.0.0 / Final Approved

1. Database Layer Models & Relations

The Mongo database is configured with 4 key collections designed with strict validations:

- User Schema: Holds authentication data (name, email, password), address entries, blocked indicator flag, wishlist references, and current sandbox wallet balance.
- Product Schema: Stores item descriptors (brand, category, brand name), variants, prices (originalPrice, localized converted INR calculations), reviews, and FAQs.
- Coupon Schema: Manages code strings, discount properties, expiration limits, and target restrictions.
- Order Schema: Records tracking status tags, item variant specs, address choice, and coupon details.

2. API Gateway Endpoints

Method	Endpoint Route	Auth Level	Description
GET	/api/products	Public	Fetch product catalog with page details
GET	/api/products/:id/related	Public	Fetch related products
POST	/api/coupons/apply	User Token	Validate coupon code discount
PUT	/api/auth/customers/:id/block	Admin Token	Toggle active customer block status
PUT	/api/orders/:id/status	Admin Token	Step forward order tracking stage

SYSTEM SOLUTION ARCHITECTURE LAYERS

PRESENTATION TIER: React SPA Client (CartContext / AuthContext / Megamenu)

LOGIC TIER: Node-Express API (Auth / Block check / Multer upload middleware)

DATA TIER: MongoDB Collections (Products / Orders) + Local JSON DB Failover Fallback

